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|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 6(a) | <p>any <b>four</b> from:</p> <ol style="list-style-type: none"> <li>1 cell wall, pathway / route / AW ;<br/> <b>R</b> crossing membranes / passing through vacuoles / passing through cytoplasm, <i>in context of passage to endodermis</i><br/> <b>I</b> diffuses / osmosis / active transport</li> <li>2 movement (also), through intercellular spaces / in spaces between cells ;<br/> <b>R</b> intracellular spaces</li> <li>3 named cell layer in <u>apoplast</u> pathway to endodermis ;<br/> e.g. epidermal cells / root hair cells<br/> cells of cortex / cortical cells / parenchyma cells<br/> <i>must be in context of cell walls</i></li> <li>4 cannot cross / stops at, Casparian strip (of endodermal cells) ;</li> <li>5 detail ;<br/> made of, suberin<br/> <b>or</b><br/> described as a, waxy / waterproof / impermeable (layer / substance / material in cell walls)</li> <li>6 (overall) movement down a water potential gradient ;<br/> <b>R</b> ref. to osmosis / active transport<br/> <i>context of across root or from root to xylem (owing to overall root to, leaf / atmosphere, gradient)</i></li> <li>7 AVP ; e.g. non-living pathway / does not cross membranes, until the endodermis<br/> idea that water needs to pass across a cell (surface) membrane to control substances entering xylem</li> </ol> | 4 |
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| Question | Answer                                               |                                     |                      |                             | Marks |
|----------|------------------------------------------------------|-------------------------------------|----------------------|-----------------------------|-------|
| 6(b)     | net movement of K <sup>+</sup>                       | membrane protein needed (yes or no) | ATP used (yes or no) | name of transport mechanism | 3     |
|          | against the concentration gradient                   | yes                                 | yes                  | active, transport / uptake  |       |
|          | down the concentration gradient                      | yes                                 | no                   | facilitated diffusion       |       |
|          | one mark each correct column ;;;                     |                                     |                      |                             |       |
|          | if no marks gained, allow one mark for a correct row |                                     |                      |                             |       |